Fiscal Year:	FY 2019	**Task Last Updated:**	FY 04/16/2019
PI Name:	Cooper, Maya M.S.		
Project Title:	Improvement of Shelf Life for Space Food Through Hurdle Approach		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	maya.cooper@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	281.483.1892
Organization Name:	Leidos/NASA Johnson Space Center		
PI Address 1:	1300 Hercules MC:C09		
PI Address 2:			
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	03/08/2017	**End Date:**	09/30/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
Contact Email:	Peter.norsk@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2019 (previously 9/30/2024) due to PI change from Maya Cooper to Takiyah Sirmons in October 2019; project continues as "Improvement of Shelf Life for Space Food Through Hurdle Approach (PI=Sirmons)" (Ed., 12/13/19)		
Key Personnel Changes/Previous PI:	January 2019 report: No personnel changes during reporting period.		
COI Name (Institution):	Sirmons, Takiyah Ph.D. (Leidos/NASA Johnson Space Center) Froio-Blumsack, Danielle M.S. (U.S. Army Natick Soldier RD&E Center) Mohr, Lea B.S. (Ameriqual) Douglas, Grace Ph.D. (NASA Johnson Space Center) Young, Millennia Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Most items of the current space food system will not achieve the minimum 5-year shelf life required for a Mars mission due to decrements in nutritional quality or sensory acceptability. Previous Advanced Food Technology (AFT) studies have shown critical losses in some nutrients in a number of space food products after 3 years of ambient storage (Cooper project, “Effects of Processing and Subsequent Storage on Nutrition (PI Cooper)”), unacceptable losses in quality after 3 years (Catauro, P.M. & Perchonok, M.H. Assessment of the long-term stability of retort pouch foods to support extended duration spaceflight. Journal of food science (2012) 77, S29-39), and the inability of individual processing and storage solutions to achieve a projected 5-year shelf life (Cooper project “Integration of Product, Package, Process, and Environment: A Food System Optimization”). This task will investigate the use of hurdle approach to increase the shelf life of the space food system. The study will produce the 5-year data essential to fully inform the state of a Mars food system and indicate the best countermeasures to nutritional and sensory degradation. Putting a 7-year data option in place initially will ensure that the PRR is not unnecessarily extended by several years if it is determined at that time that a longer-term shelf life study was necessary to determine mission requirements. Specific Aims: Determine how reduced storage temperatures (-80C, -20C, 4C) and alternative processing and packaging impact the quality and nutrient concentrations of space food over a 5-7 year shelf life period as compared to the quality and nutrient concentrations of space food produced under traditional methods and stored at ambient temperature (21C).
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research, which requires focused and constrained data gathering and analysis that is more appropriately obtained through a non-competitive proposal. The results of this study will influence vehicle design. We have established a collaboration with the Advanced Exploration Systems Logistics Reduction (AES LR) team, who will contribute their own funding to evaluate storage solutions and feasibility of designs. AES will ultimately be responsible for the food storage solution on a mission to Mars, and their location at Johnson Space Center and their collaboration at this initial point will ensure that solutions are integrated successfully in the final vehicle design. Only four storage conditions can be assessed within this experiment, due to funding limitations. The AES team participated in discussions in determining the experimental storage conditions to ensure direct applicability to potential scenarios of food storage for long duration missions. The current AES work in vehicle design, and their collaboration with Advanced Food Technology to finalize storage solutions, is the first reason that we propose that this work be accomplished through directed study.
Research Impact/Earth Benefits:	The discovery of pathways to extend the shelf life of space food is directly transferable to the extension of shelf life of military food, emergency food rations, and commercial food items. Longer food shelf life leads to lower food waste, higher food quality in aging food samples, and an opportunity to increase variety within stored food inventory. The exploration of an alternative technology, specifically microwave-assisted thermal sterilization, provides additional data for the FDA (Food & Drug Administration) approval of this new processing technology, which drives commercial innovation in the food industry.
Task Progress:	Sixteen representative foods from the International Space Station food system were chosen for production and analysis and will be evaluated initially and at 1, 3, and 5 years, with potential for analysis at 7 years if necessary. Analysis includes changes in color, texture, nutrition, sensory quality, and rehydration ratio when applicable. The food samples are stored at -20°C, 4°C, and 21°C. Select food samples will also be evaluated at -80°C to determine the impacts of ultra-cold storage after 1 and 5 years. Eight condiments were added to the study this year for quality evaluation at the same time intervals. Packaging film barrier properties and mechanical integrity was assessed before and after processing and will be assessed after storage as well. At the study conclusion, if tested hurdles are adequate, formulation, processing, and storage combinations will be uniquely identified for processed food matrices to achieve a 5-year shelf life. Storage evaluations have begun for 12 of the 16 foods and all eight condiments. Substantial declines in vitamin A, vitamin C, vitamin E, thiamin, and folic acid concentrations were noted in some foods stored at 21°C for 1 year. Both smaller declines and undetectable changes in these same nutrients occurred at lower storage temperatures. Lycopene did not show significant differences that correlated to the food storage temperature. Additional changes were noted in color, texture, and moisture content for some of the foods after ambient storage. Overall acceptability of most foods was not different after one-year of storage independent of storage temperature. [Ed. note 12/13/2019: Project continues as "Improvement of Shelf Life for Space Food Through Hurdle Approach (PI=Sirmons)" due to PI Cooper relocating within Human Research Program at Johnson Space Center. See that project for subsequent reporting]
Bibliography Type:	Description: (Last Updated: 04/23/2019)
Abstracts for Journals and Proceedings	Cooper MR, Sirmons TA, Froio-Blumsack D, Young M, Douglas GL. "Extension of Space Food Shelf Life Through Hurdle Approach." Poster presentation at the 2019 NASA Human Research Program Investigators’ Workshop, Galveston, TX, January 22-25, 2019. 2019 NASA Human Research Program Investigators’ Workshop, Galveston, TX, January 22-25, 2019. Available online: http://cdn-uploads.preciscentral.com/Download/Submissions/D7DB3F75E6939F5F/E1ECE8C166AD99E4.pdf ; accessed 4/23/19. , Jan-2019